



## Q 1. What is SQL?

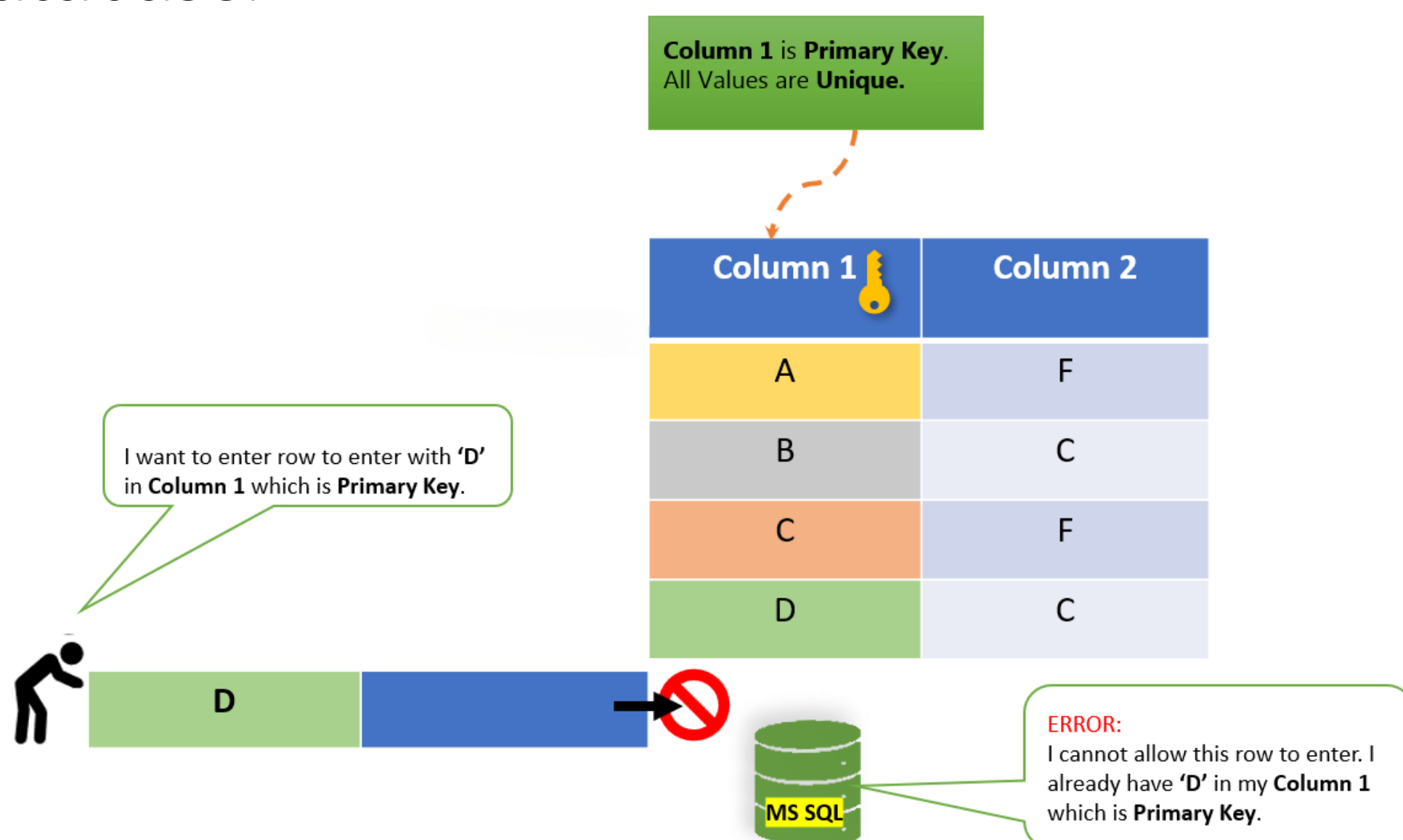
**Ans:** SQL stands for Structured Query Language. It is a programming language used for managing and manipulating relational databases.

## Q 2. What is a database?

**Ans:** A database is an organized collection of data stored and accessed electronically. It provides a way to store, organize, and retrieve large amounts of data efficiently.

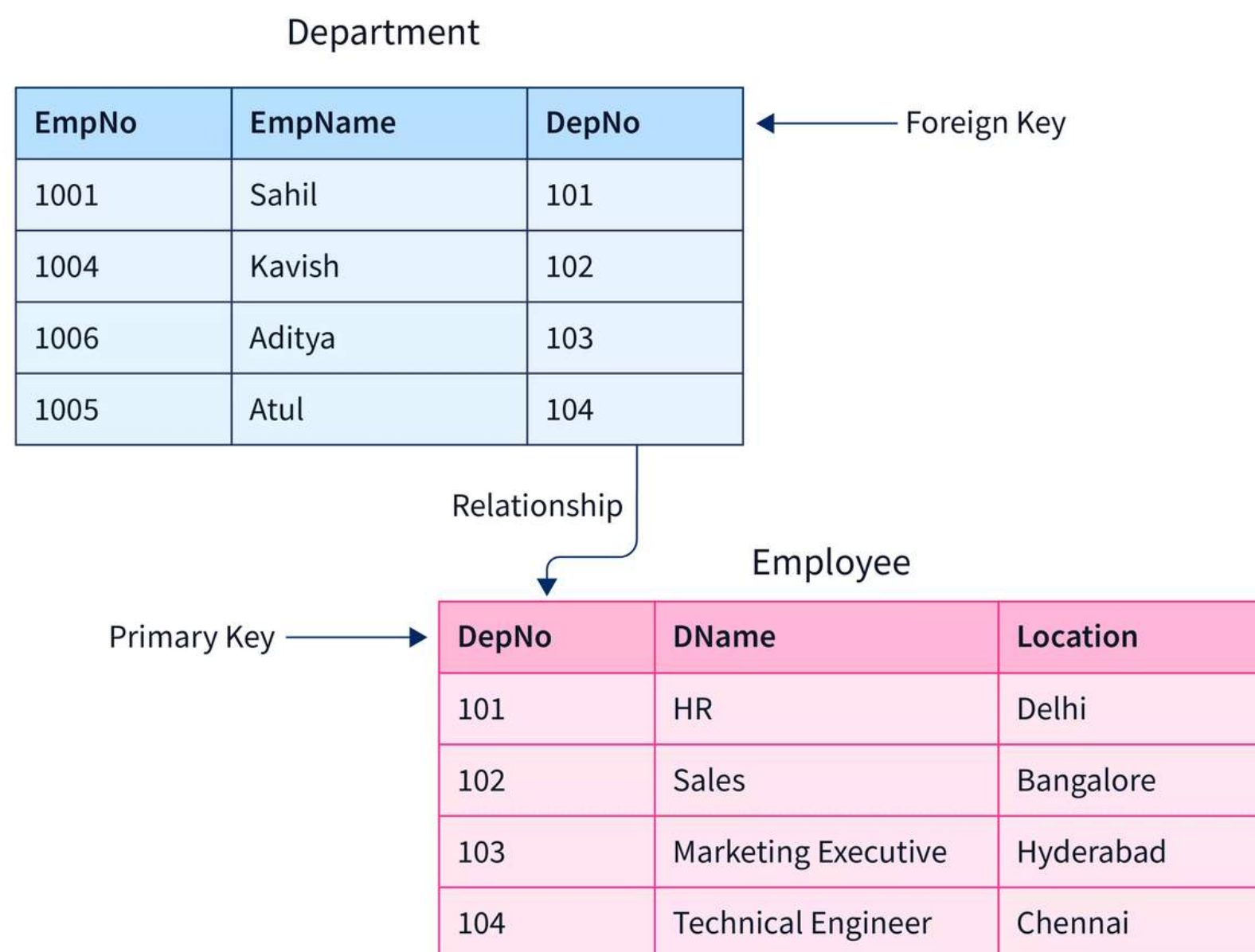
## Q 3. What is a primary key?

**Ans:** A primary key is a column or combination of columns that uniquely identifies each row in a table. It enforces the entity integrity rule in a relational database.



## Q 4. What is a foreign key?

**Ans:** A foreign key is a column or combination of columns that establishes a link between data in two tables. It ensures referential integrity by enforcing relationships between tables.



## Q 5. What is the difference between a primary key and a unique key?

**Ans:** A primary key is used to uniquely identify a row in a table and must have a unique value. On the other hand, a unique key ensures that a column or combination of columns has a unique value but does not necessarily identify the row.

## Q 6. What is normalization?

**Ans:** Normalization is the process of organizing data in a database to minimize redundancy and dependency. It involves breaking down a table into smaller tables and establishing relationships between them.

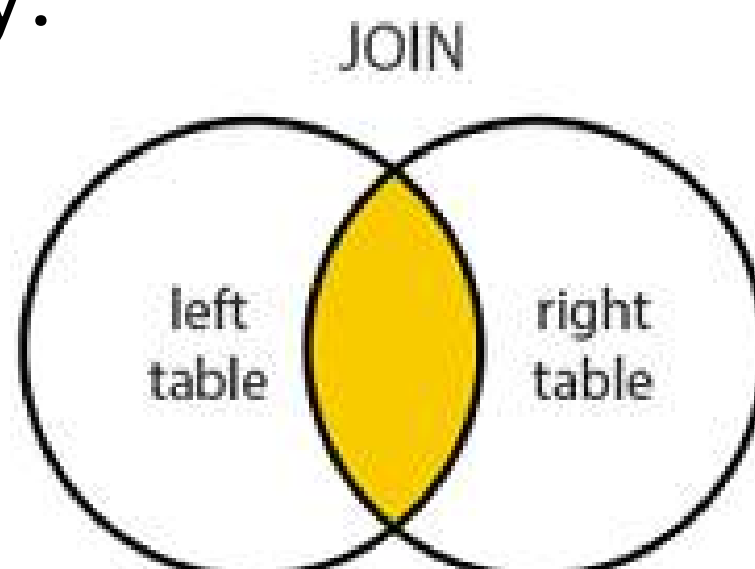
## Q 7. What are the different types of normalization?

**Ans:** The different types of normalization are:

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce-Codd Normal Form (BCNF)
- Fourth Normal Form (4NF)
- Fifth Normal Form (5NF) or Project-Join Normal Form (PJNF)

## Q 8. What is a join in SQL?

**Ans:** A join is an operation used to combine rows from two or more tables based on related columns. It allows you to retrieve data from multiple tables simultaneously.



## Q 9. What is the difference between DELETE and TRUNCATE in SQL?

**Ans:** The **DELETE statement** is used to remove specific rows from a table based on a condition. It can be rolled back and generates individual delete operations for each row.

**TRUNCATE**, on the other hand, is used to remove all rows from a table. It cannot be rolled back, and it is faster than DELETE as it deallocates the data pages instead of logging individual row deletions.

## Q 10. What is the difference between UNION and UNION ALL?

**Ans:** UNION and UNION ALL are used to combine the result sets of two or more SELECT statements.

**UNION** removes duplicate rows from the combined result set.

whereas **UNION ALL** includes all rows, including duplicates.

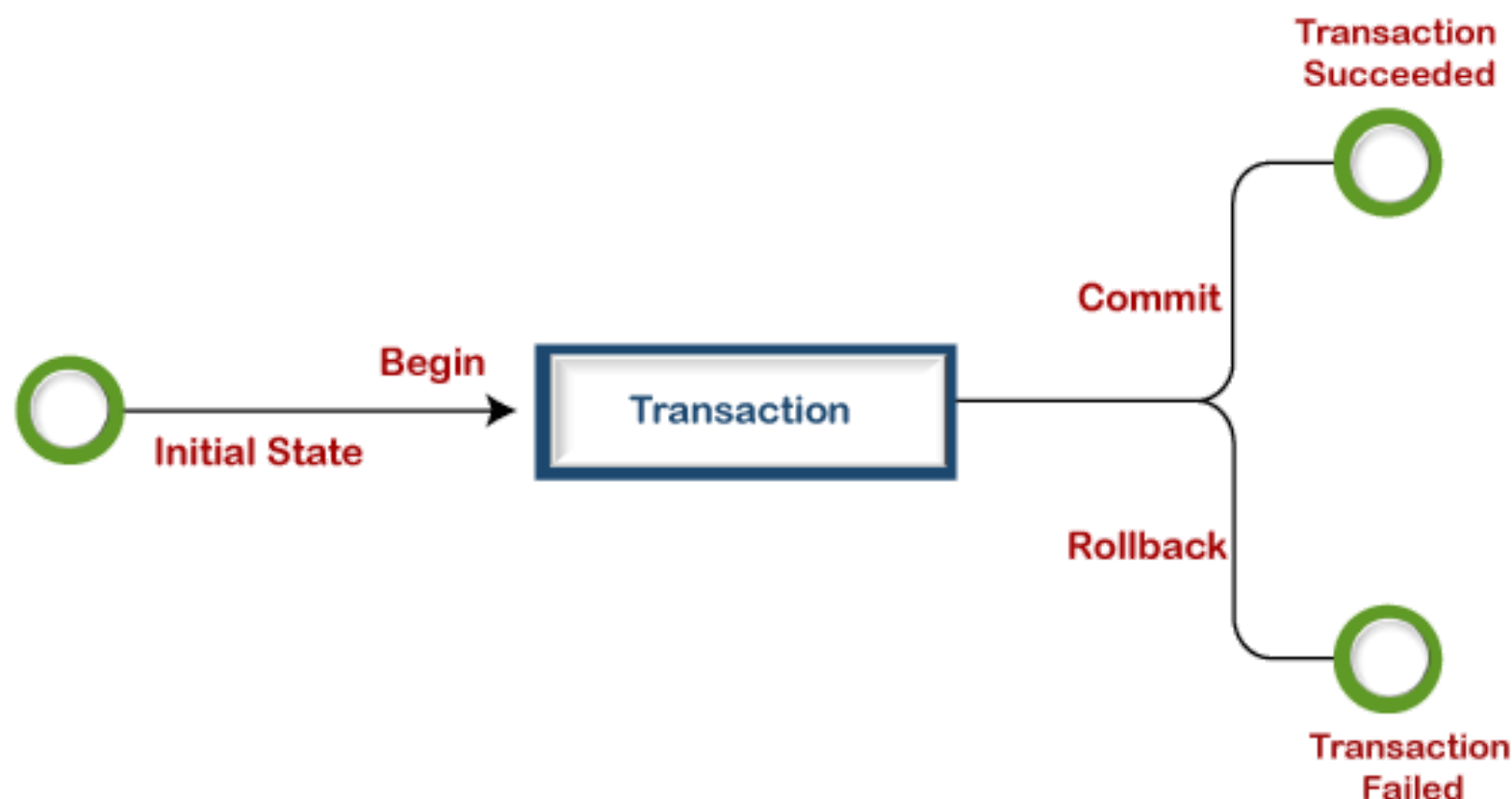
## Q 11. What is the difference between the **HAVING** clause and the **WHERE** clause?

**Ans:** The **WHERE** clause is used to filter rows based on a condition before the data is grouped or aggregated. It operates on individual rows.

The **HAVING** clause, on the other hand, is used to filter grouped rows based on a condition after the data is grouped or aggregated using the **GROUP BY** clause.

## Q 12. What is a transaction in SQL?

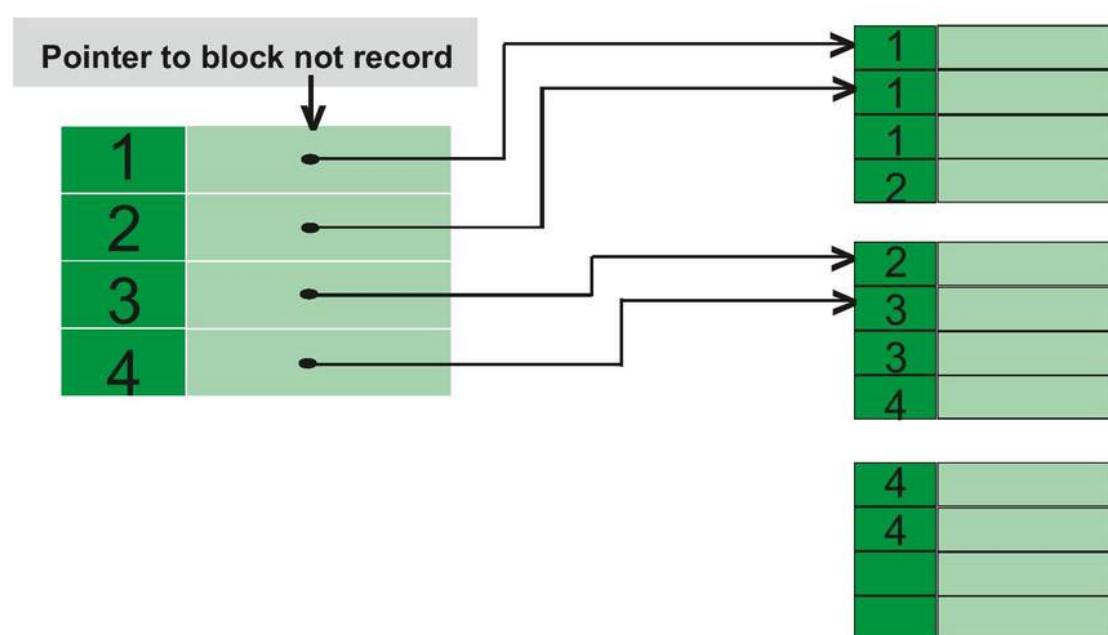
**Ans:** A transaction is a sequence of SQL statements that are executed as a single logical unit of work. It ensures data consistency and integrity by either committing all changes or rolling them back if an error occurs.



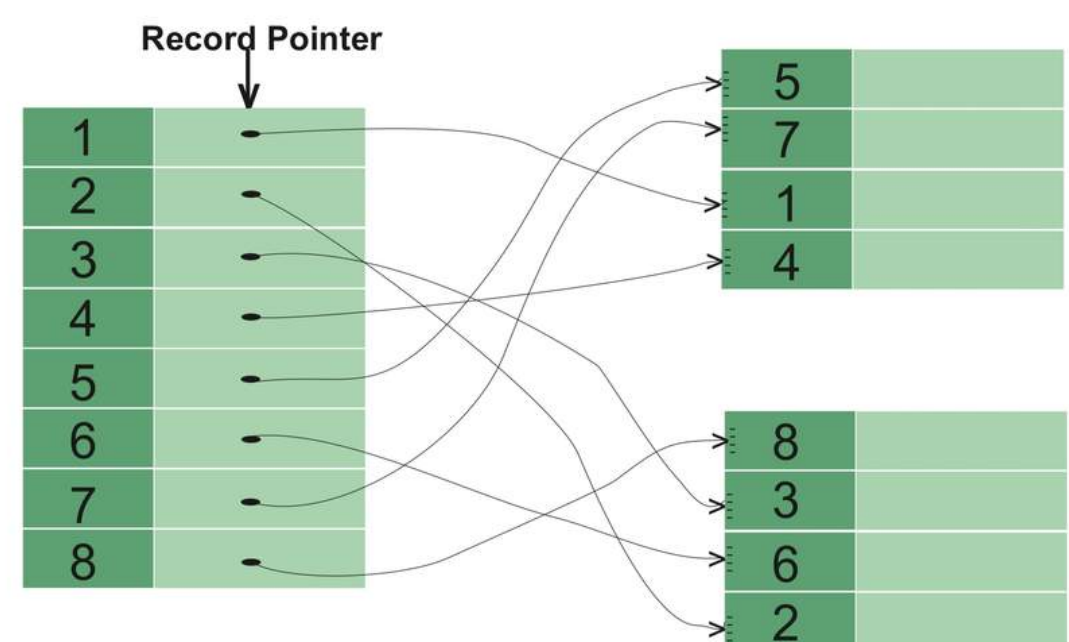
## Q 13. What is the difference between a clustered and a non-clustered index?

**Ans:** A **clustered index** determines the physical order of data in a table. It changes the way the data is stored on disk and can be created on only one column. A table can have only one clustered index.

A **non-clustered index** does not affect the physical order of data in a table. It is stored separately and contains a pointer to the actual data. A table can have multiple non-clustered indexes.



**Clustered Index**



**Non-Clustered Index**



## Q 14. What is ACID in the context of database transactions?

**Ans:** ACID stands for Atomicity, Consistency, Isolation, and Durability. It is a set of properties that guarantee reliable processing of database transactions.

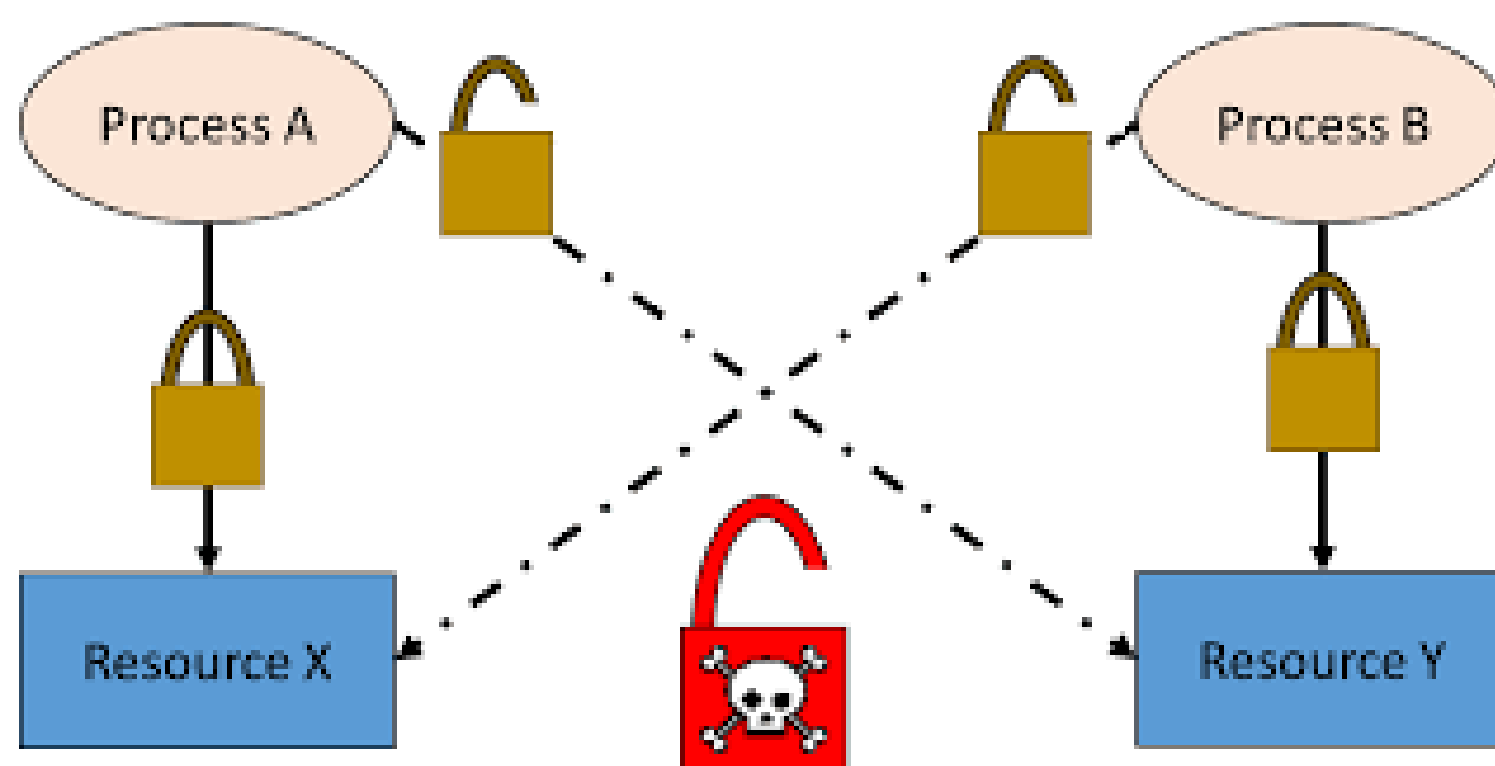
- **Atomicity** ensures that a transaction is treated as a single unit of work, either all or none of the changes are applied.
- **Consistency** ensures that a transaction brings the database from one valid state to another.
- **Isolation** ensures that concurrent transactions do not interfere with each other.
- **Durability** ensures that once a transaction is committed, its changes are permanent and survive system failures.

**A** - Atomicity  
**C** - Consistency  
**I** - Isolation  
**D** - Durability



## Q 15. What is a deadlock?

**Ans:** A deadlock occurs when two or more transactions are waiting for each other to release resources, resulting in a circular dependency. As a result, none of the transactions can proceed, and the system may become unresponsive.



## Q 16. What is the difference between a database and a schema?

**Ans:** A database is a container that holds multiple objects, such as tables, views, indexes, and procedures. It represents a logical grouping of related data.

A schema, on the other hand, is a container within a database that holds objects and defines their ownership. It provides a way to organize and manage database objects.

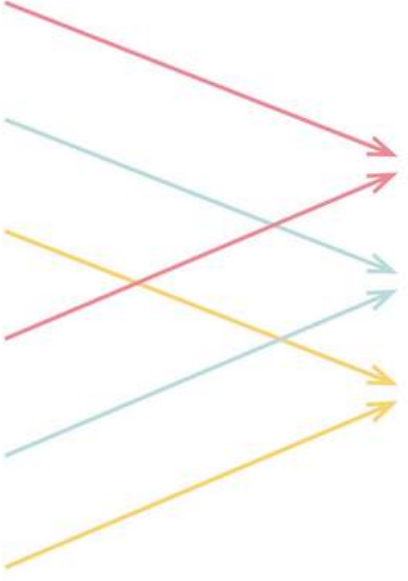
## Q 17. What is the difference between a temporary table and a table variable?

**Ans:** A temporary table is a table that is created and exists only for the duration of a session or a transaction. It can be explicitly dropped or is automatically dropped when the session or transaction ends.

A table variable is a variable that can store a table-like structure in memory. It has a limited scope within a batch, stored procedure, or function. It is automatically deallocated when the scope ends.

## Q 18. What is the purpose of the GROUP BY clause?

**Ans:** The GROUP BY clause is used to group rows based on one or more columns in a table. It is typically used in conjunction with aggregate functions, such as SUM, AVG, COUNT, etc., to perform calculations on grouped data.



| title  | genre     | qty |
|--------|-----------|-----|
| book 1 | adventure | 4   |
| book 2 | fantasy   | 5   |
| book 3 | romance   | 2   |
| book 4 | adventure | 3   |
| book 5 | fantasy   | 3   |
| book 6 | romance   | 1   |

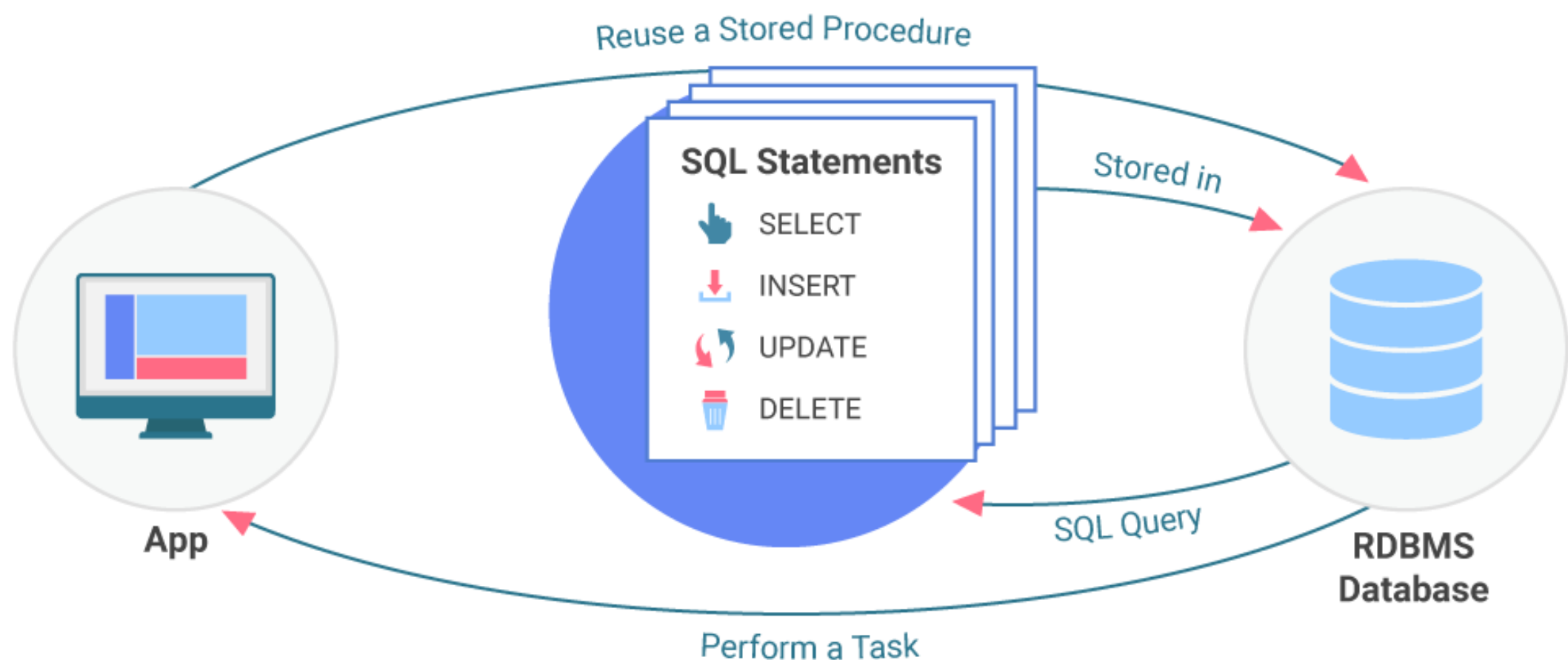
| genre     | total |
|-----------|-------|
| adventure | 7     |
| fantasy   | 8     |
| romance   | 3     |

## Q 19. What is the difference between CHAR and VARCHAR data types?

**Ans:** CHAR is a fixed-length string data type, while VARCHAR is a variable-length string data type.

## Q 20. What is a stored procedure?

**Ans:** A stored procedure is a set of SQL statements that are stored in the database and can be executed repeatedly. It provides code reusability and better performance.



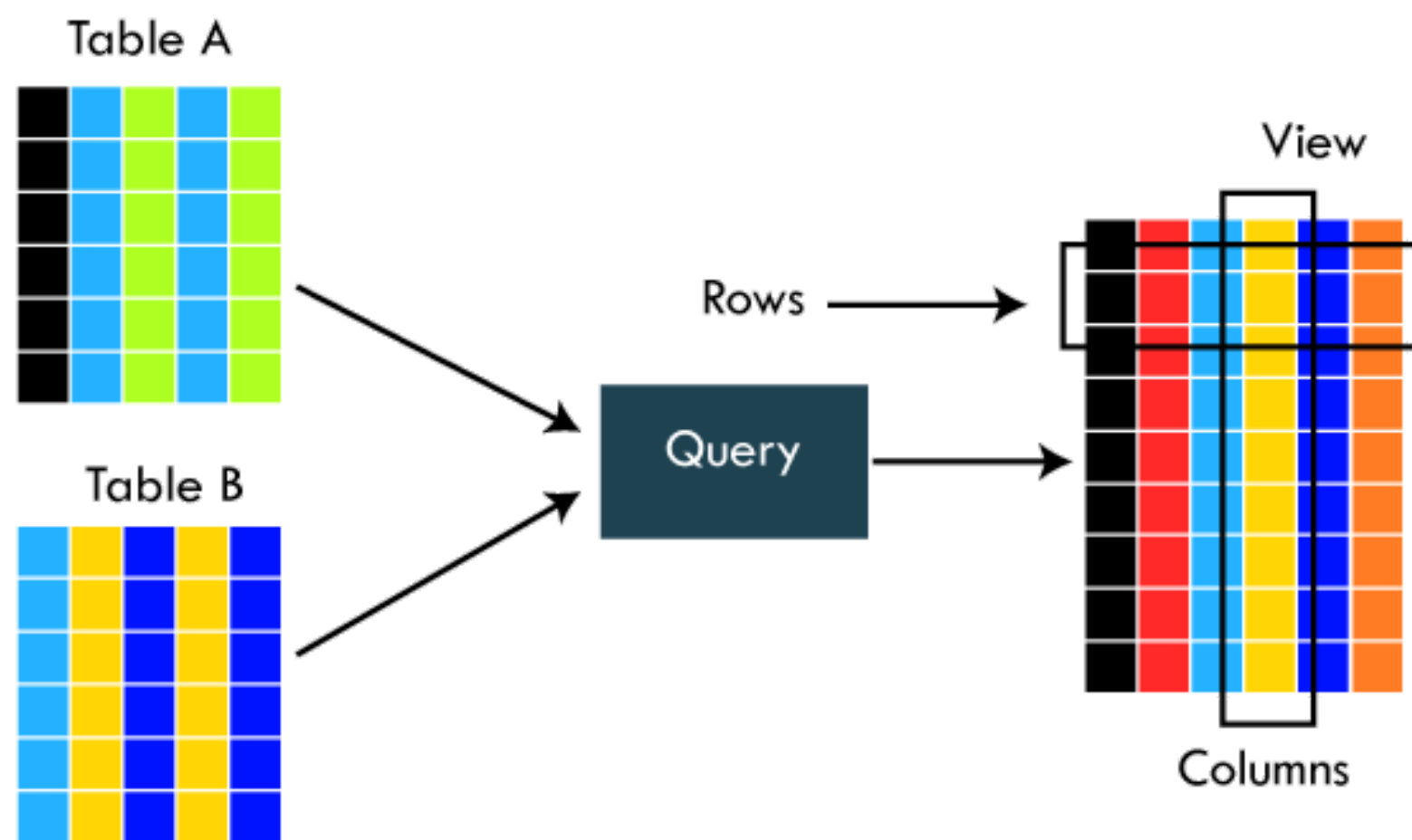
## Q 21. What is a subquery?

**Ans:** A subquery is a query nested inside another query. It is used to retrieve data based on the result of an inner query.



## Q 22. What is a view?

**Ans:** A view is a virtual table based on the result of an SQL statement. It allows users to retrieve and manipulate data as if





**Q 23. What is the difference between a cross join and an inner join?**

**Ans:** A cross join (Cartesian product) returns the combination of all rows from two or more tables.

An inner join returns only the matching rows based on a join condition.

**Q 24. What is the purpose of the COMMIT statement?**

**Ans:** The COMMIT statement is used to save changes made in a transaction permanently. It ends the transaction and makes the changes visible to other users.

**Q 25. What is the purpose of the ROLLBACK statement?**

**Ans:** The ROLLBACK statement is used to undo changes made in a transaction. It reverts the database to its previous state before the transaction started.

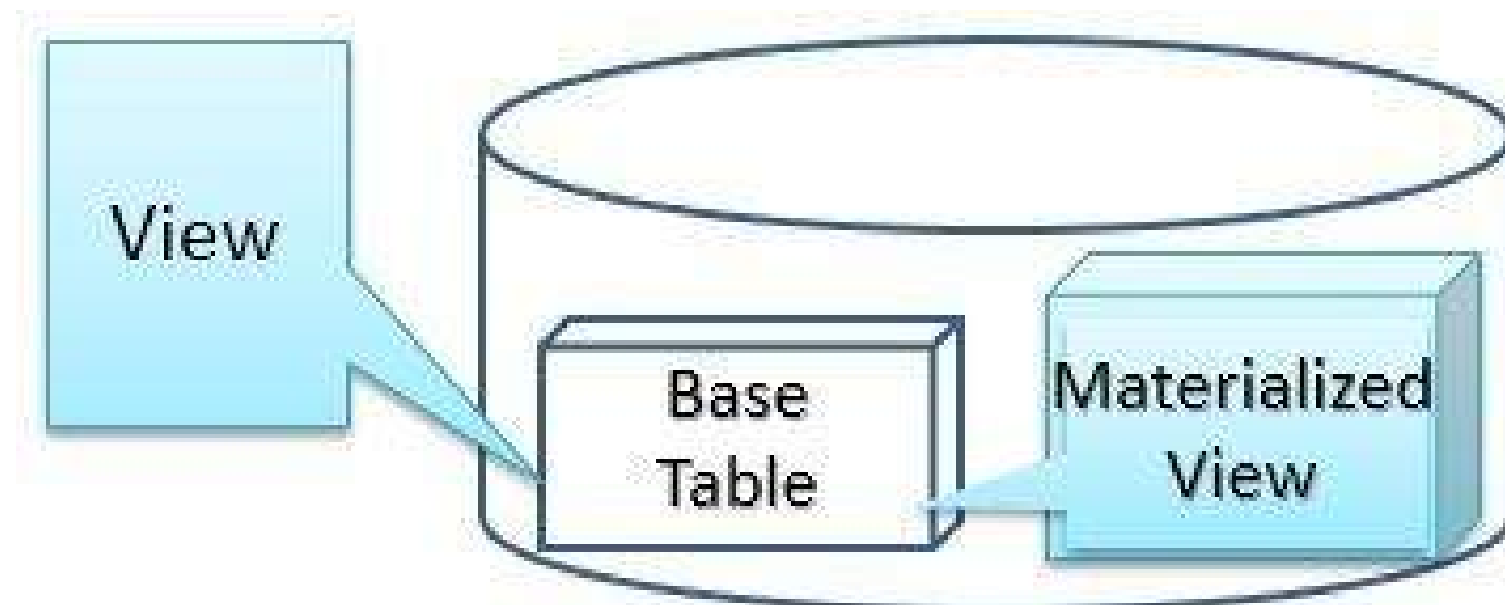


**Q 26. What is the purpose of the NULL value in SQL?**

**Ans:** NULL represents the absence of a value or unknown value. It is different from zero or an empty string and requires special handling in SQL queries.

**Q 27. What is the difference between a view and a materialized view?**

**Ans:** A materialized view is a physical copy of the view's result set stored in the database, which is updated periodically. It improves query performance at the cost of data freshness.





## Q 28. What is a correlated subquery?

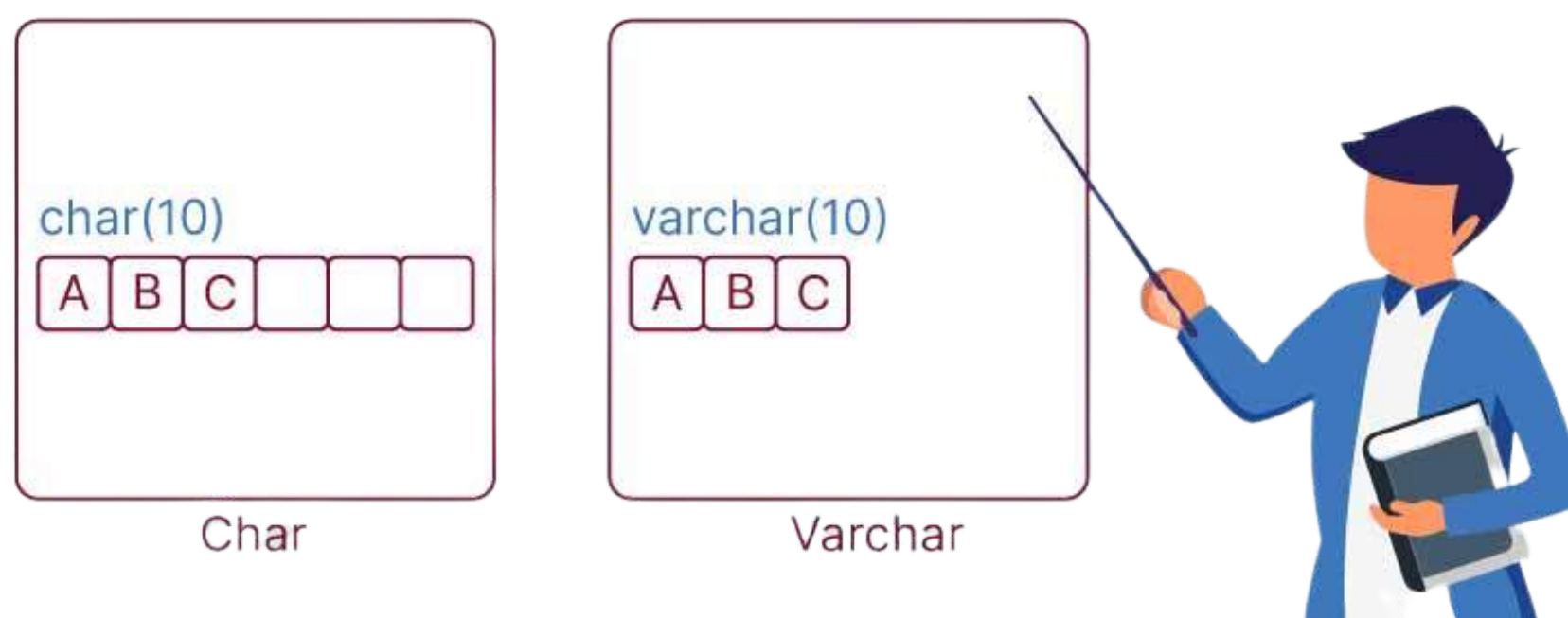
**Ans:** A correlated subquery is a subquery that refers to a column from the outer query. It executes once for each row processed by the outer query.

## Q 29. What is the purpose of the DISTINCT keyword?

**Ans:** The DISTINCT keyword is used to retrieve unique values from a column or combination of columns in a SELECT statement.

## Q 30. What is the difference between the CHAR and VARCHAR data types?

**Ans:** CHAR stores fixed-length character strings, while VARCHAR stores variable-length character strings. The storage size of CHAR is constant, while VARCHAR adjusts dynamically.





**Q 31. What is the difference between the IN and EXISTS operators?**

**Ans:** The IN operator checks for a value within a set of values or the result of a subquery. The EXISTS operator checks for the existence of rows returned by a subquery.

**Q 32. What is the purpose of the TRIGGER statement?**

**Ans:** The TRIGGER statement is used to associate a set of SQL statements with a specific event in the database. It is executed automatically when the event occurs

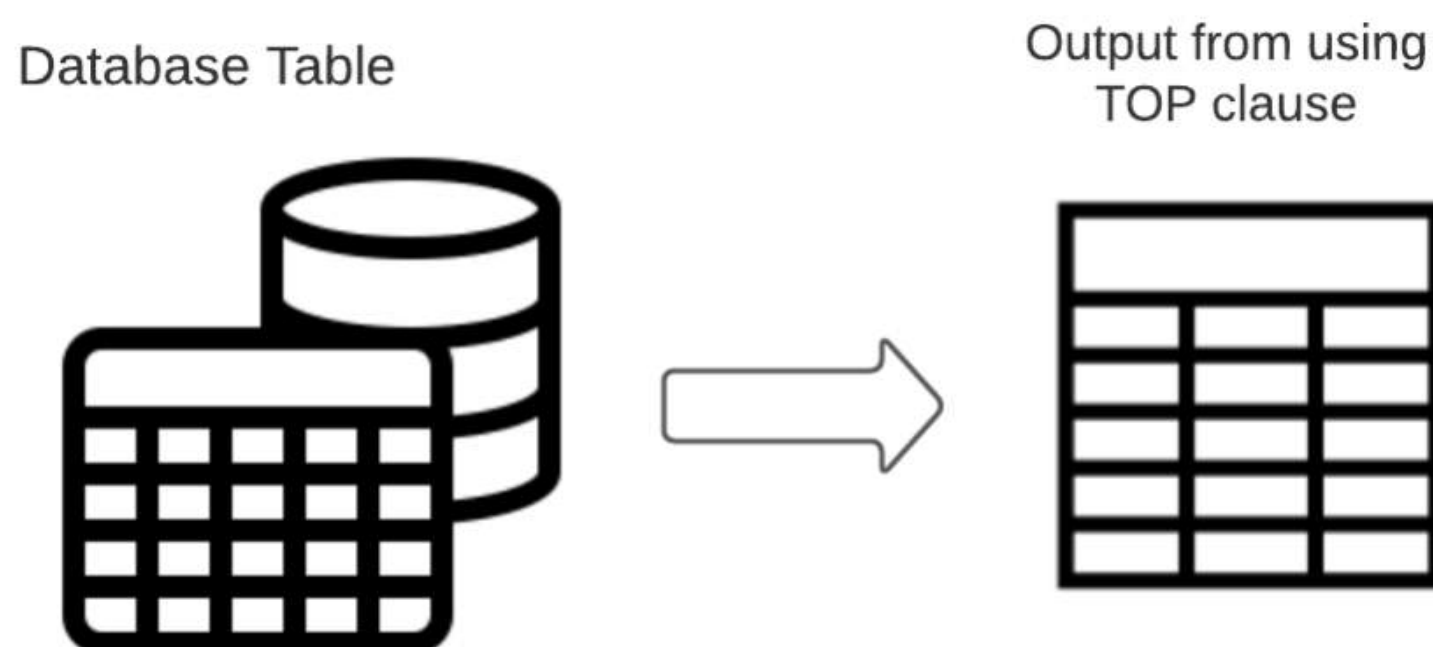
**Q 33. What is the difference between a unique constraint and a unique index?**

**Ans:** A unique constraint ensures the uniqueness of values in one or more columns, while a unique index enforces the uniqueness and also improves query performance.



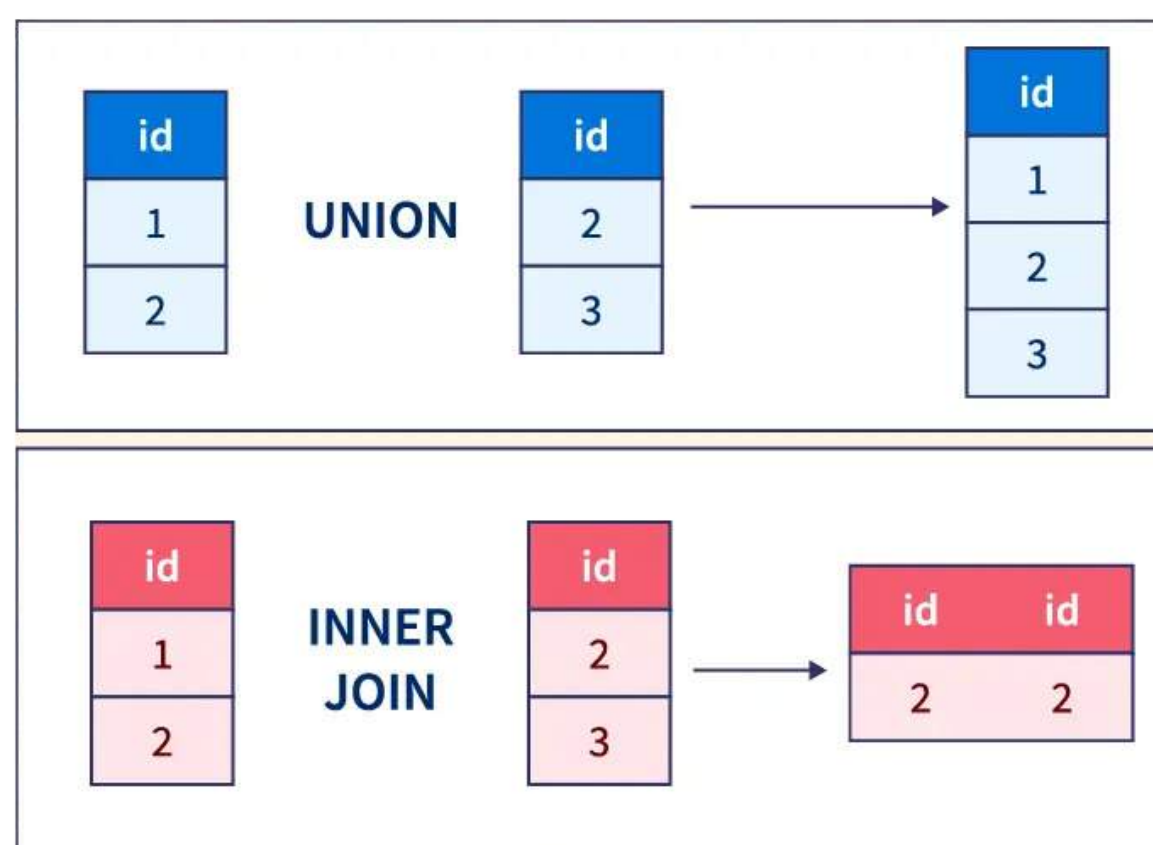
## Q 34. What is the purpose of the TOP or LIMIT clause?

**Ans:** The TOP (in SQL Server) or LIMIT (in MySQL) clause is used to limit the number of rows returned by a query. It is often used with an ORDER BY clause.



## Q 35. What is the difference between the UNION and JOIN operators?

**Ans:** UNION combines the result sets of two or more SELECT statements vertically, while JOIN combines columns from two or more tables horizontally based on a join condition.





## Q 36. What is a data warehouse?

**Ans:** A data warehouse is a large, centralized repository that stores and manages data from various sources. It is designed for efficient reporting, analysis, and business intelligence purposes.



## Q 37. What is the difference between a primary key and a candidate key?

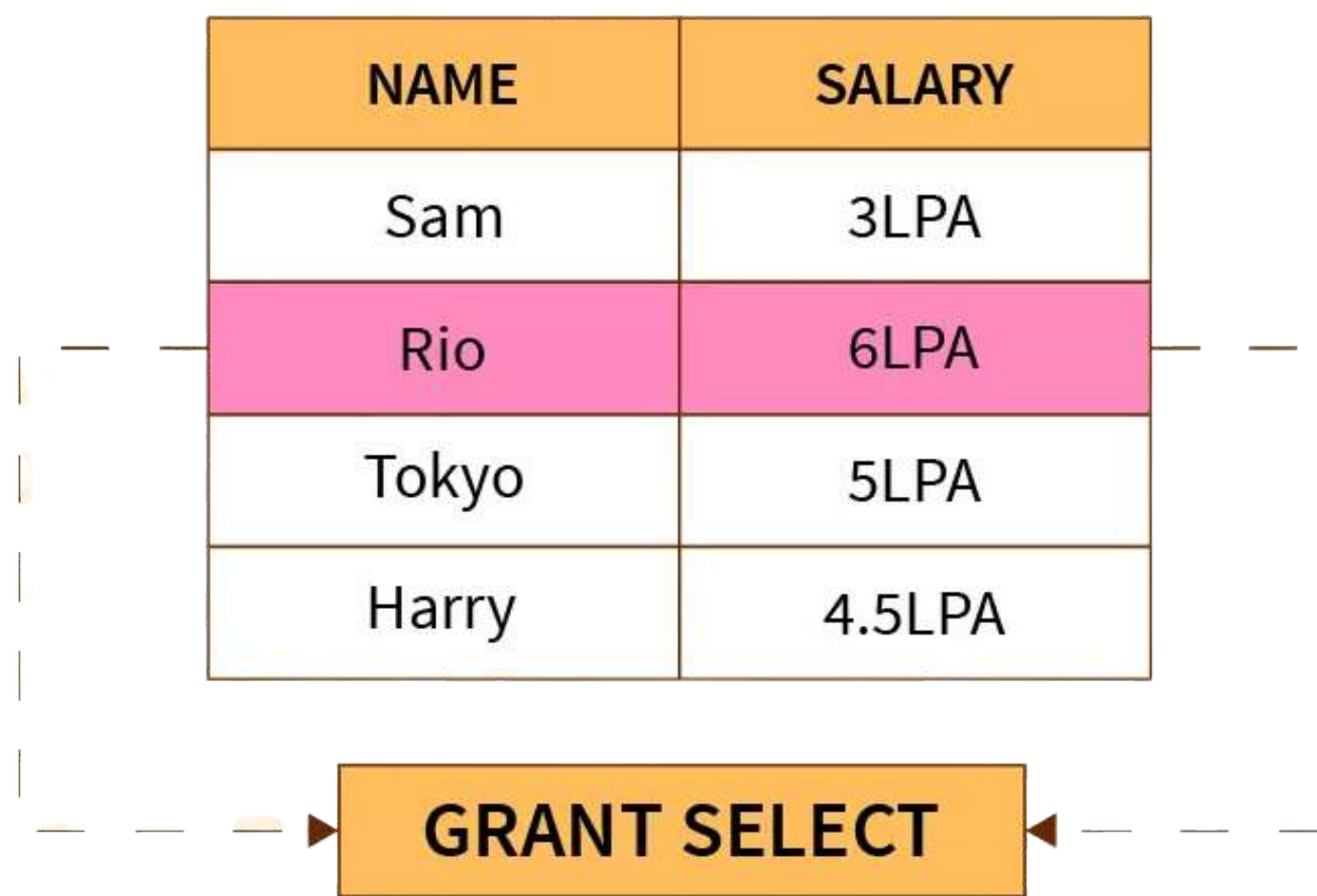
**Ans:** A primary key is a chosen candidate key that uniquely identifies a row in a table.

A candidate key is a set of one or more columns that could potentially become the primary key.



### Q 38. What is the purpose of the GRANT statement?

**Ans:** The GRANT statement is used to grant specific permissions or privileges to users or roles in a database.



### Q 39. What is a correlated update?

**Ans:** A correlated update is an update statement that refers to a column from the same table in a subquery. It updates values based on the result of the subquery for each row.



## Q 40. What is the purpose of the CASE statement?

**Ans:** The CASE statement is used to perform conditional logic in SQL queries. It allows you to return different values based on specified conditions.

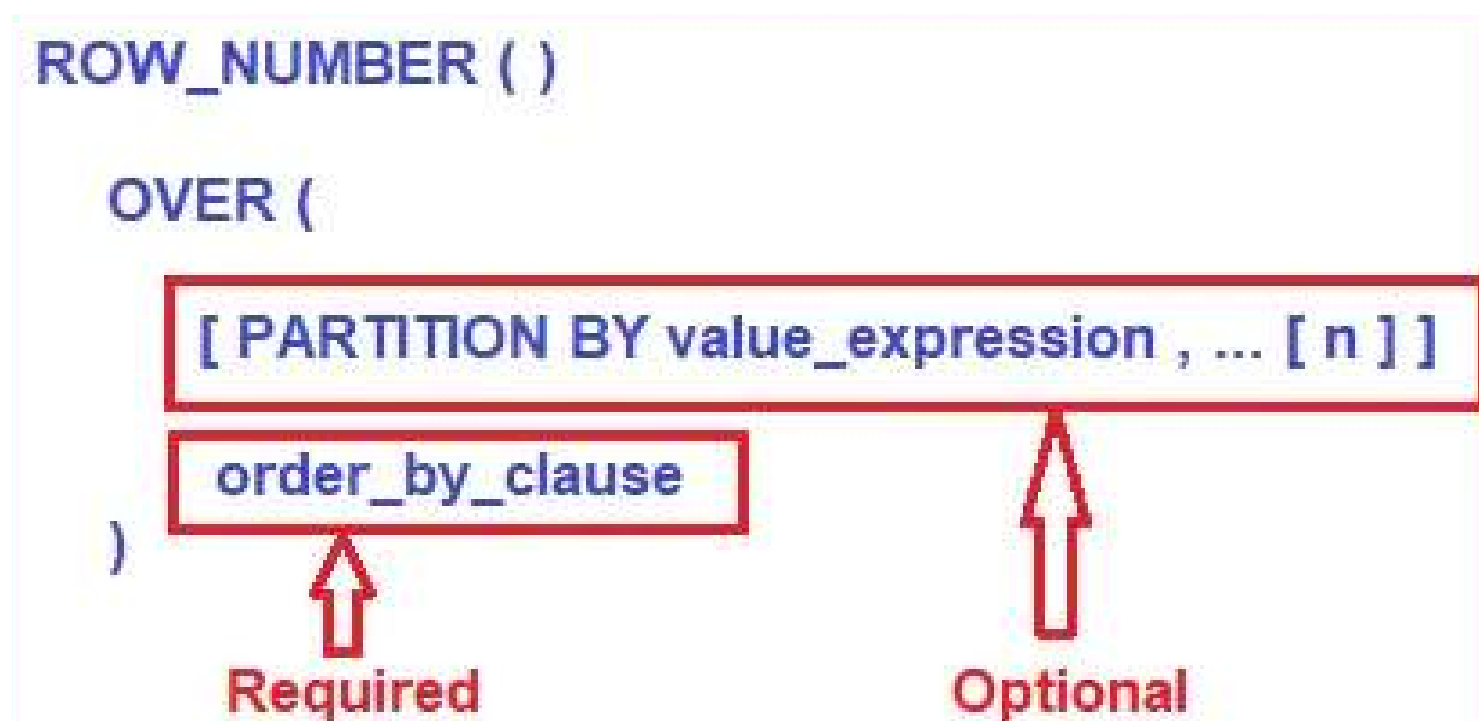
## Q 41. What is the purpose of the COALESCE function?

**Ans:** The COALESCE function returns the first non-null expression from a list of expressions. It is often used to handle null values effectively.

## Q 42. What is the purpose of the ROW\_NUMBER() function?

**Ans:** The ROW\_NUMBER() function assigns a unique incremental number to each row in the result set.

It is commonly used for pagination or ranking purposes.



**Q 43. What is the difference between a natural join and an inner join?**

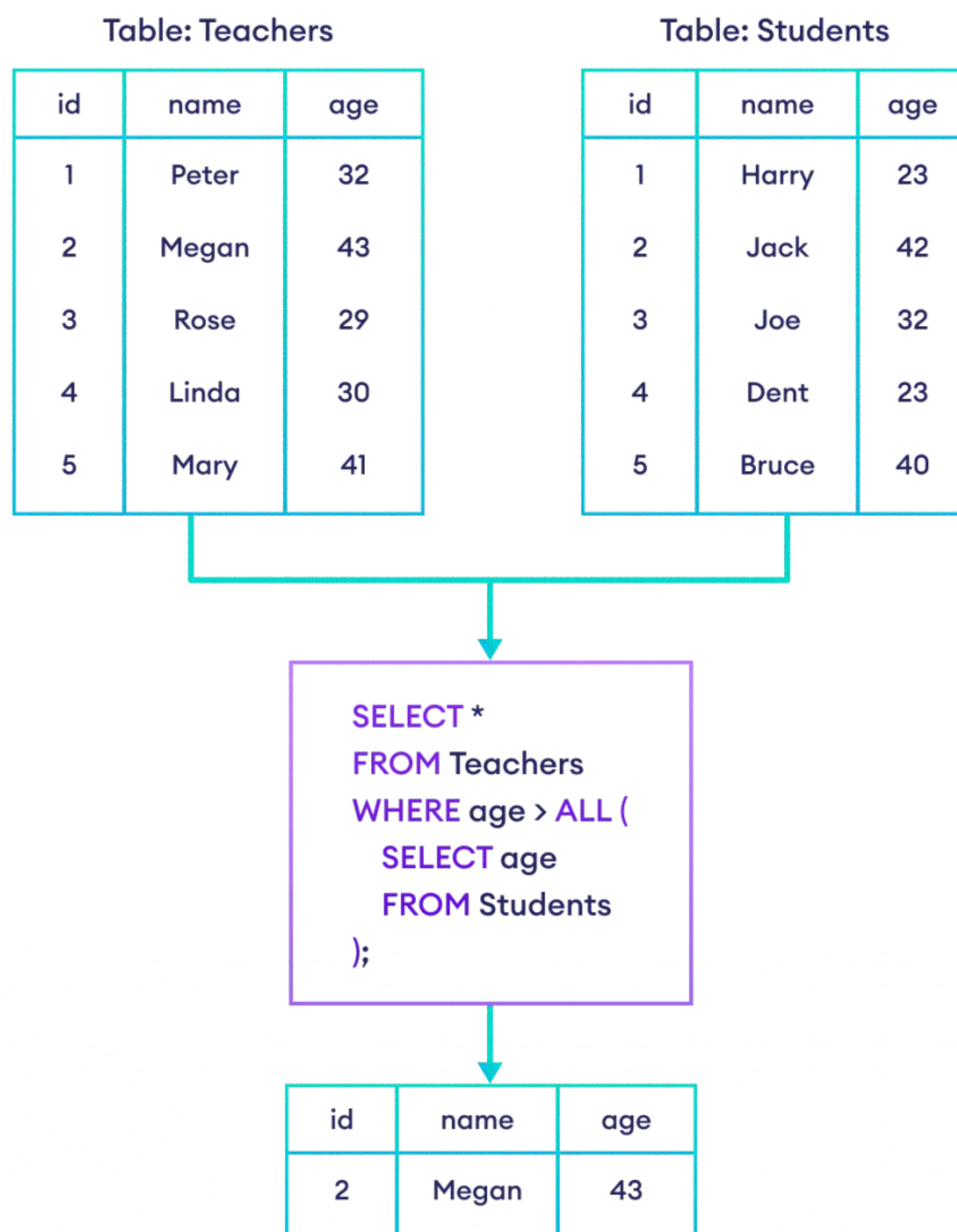
**Ans:** A natural join is an inner join that matches rows based on columns with the same name in the joined tables. It is automatically determined by the database.

**Q 44. What is the purpose of the CASCADE DELETE constraint?**

**Ans:** The CASCADE DELETE constraint is used to automatically delete related rows in child tables when a row in the parent table is deleted.

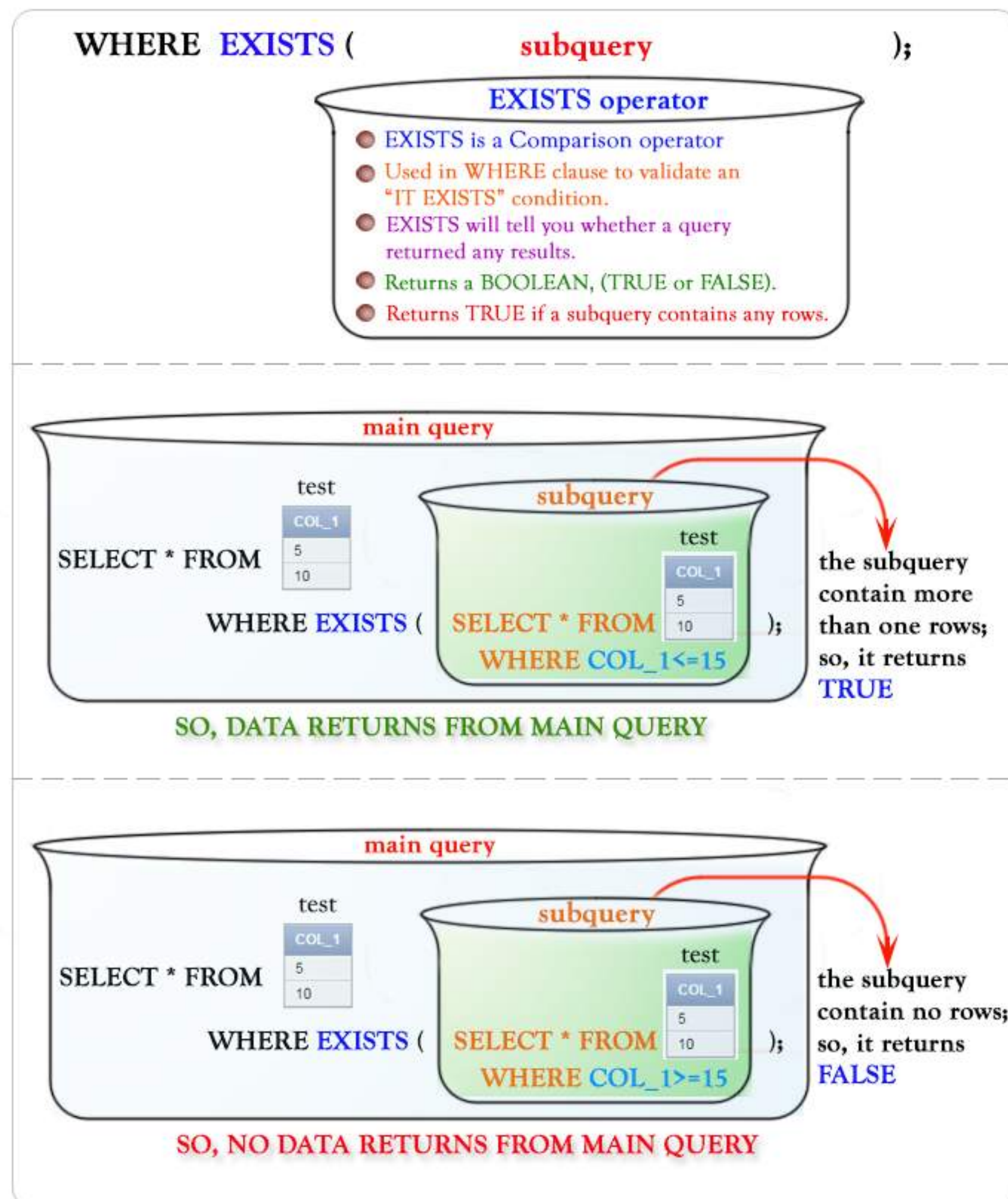
## Q 45. What is the purpose of the ALL keyword in SQL?

**Ans:** The CASCADE DELETE constraint is used to automatically delete related rows in child tables when a row in the parent table is deleted.



## Q 46. What is the difference between the EXISTS and NOT EXISTS operators?

**Ans:** The EXISTS operator returns true if a subquery returns any rows, while the NOT EXISTS operator returns true if a subquery returns no rows.



## Q 47. What is the purpose of the CROSS APPLY operator?

**Ans:** The CROSS APPLY operator is used to invoke a table-valued function for each row of a table expression. It returns the combined result set.

## Q 48. What is a self-join?

**Ans:** A self-join is a join operation where a table is joined with itself. It is useful when you want to compare rows within the same table based on related columns. It requires a combined result set.

| Name  | Age | Location    | ID    |
|-------|-----|-------------|-------|
| John  | 35  | California  | 12698 |
| Harry | 24  | Los Angeles | 12699 |
| Smith | 32  | Arizona     | 12700 |
| Gary  | 45  | New Jersey  | 12701 |

| Ref ID | Name  | Age | Location    |
|--------|-------|-----|-------------|
| 12701  | Gary  | 45  | New Jersey  |
| 12699  | Harry | 24  | Los Angeles |
| 12698  | Smith | 32  | Arizona     |
| 12700  | Gary  | 45  | New Jersey  |

## Q 49. What is an ALIAS command?

**Ans:** ALIAS command in SQL is the name that can be given to any table or a column. This alias name can be referred in WHERE clause to identify a particular table or a column.

## Q 50. Why are SQL functions used?

**Ans:** SQL functions are used for the following purposes:

- To perform some calculations on the data
- To modify individual data items
- To manipulate the output
- To format dates and numbers
- To convert the data types

### SQL Functions

